Profile-Driven Hardware Simulation for Organic Optoelectronic Edge Vision

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Abstract

Organic optoelectronic devices offer multilevel conductance tuning and low static power, yet whether these characteristics suffice for modern vision tasks remains unknown. We present a profile-driven first-order behavioral simulation framework that maps literature-derived device metrics to task-level performance through a replaceable profile interface, incorporating photoresponse, retention, and write nonlinearity that are not directly captured by inorganic-memory simulators.

Across Tiny-ViT-5M, ConvNeXt-Tiny, and ResNet-18 on three vision datasets, we find that ADC resolution—not nominal conductance quantization—dominates accuracy under the simulated canonical regime: reducing ADC resolution below 6 bits causes abrupt accuracy collapse, and a Sobol decomposition over a 63-point D2D–ADC grid confirms that once 6-bit readout is secured, device-to-device variability dominates the residual accuracy budget. Standard HAT overfits a single D2D realization and collapses to 10.00% on fresh hardware, whereas Ensemble HAT recovers $86.37 \pm 1.54\%$ by resampling D2D masks during training. A literature-anchored 2025 OPECT case study reaches 88.53% zero-shot transfer accuracy in the same workflow. Under the present gradient-scaling approximation, however, severe nonlinear write ($NL = 2.0$) presents a practical recovery bottleneck for the current training recipe and limits accuracy to $27.72 \pm 0.82\%$.

Taken together, the framework serves as a simulation-based materials-to-system decision aid for identifying device characteristics that constrain edge-vision deployment.

1 Introduction

The energy cost of moving data between memory and arithmetic units motivates compute-in-memory (CIM) and processing-in-memory (PIM) architectures for edge inference[1;2]. In these architectures, dense matrix-vector multiplications execute directly within memory arrays, eliminating the von Neumann bottleneck that limits conventional digital systems. CIM implementations span diverse technologies: conventional SRAM and DRAM arrays, emerging resistive memories (RRAM, PCM), and organic optoelectronic devices that combine optical sensitivity with synaptic plasticity[3;4].

Organic optoelectronic and electrochemical synaptic devices are particularly attractive for edge vision applications because of their multilevel tunability, low-voltage operation, compatibility with flexible substrates, and inherent photosensitivity that enables direct optical signal processing[3–5]. Recent demonstrations include multilevel memory with analog conductance states[4;6], long retention tails suitable for inference workloads[7], and integrated optoelectronic synaptic arrays for visual computing[8;9].

However, translating these device-level advances into system-level vision performance faces a critical methodological gap. The organic optoelectronic literature remains dominated by device-centric characterizations. Network-level studies are typically limited to small benchmarks or simple perceptrons, while variability, ADC resolution, retention-induced drift, and transfer across fabricated instances are either absent from system analyses or treated only qualitatively[6;7]. The most relevant precursor for variability-aware organic simulation remains the Monte Carlo study of Alibart *et al.*, which sampled device parameters for a simple organic-memristive perceptron[10].

No clear framework yet connects reported organic-device characteristics to deployment-facing accuracy expectations for modern vision transformers or CNNs.

This gap is particularly acute for hardware-aware training (HAT), which mitigates analog non-idealities by injecting device noise into the forward training path [11]. Although related in spirit to quantization-aware training [12] and to domain randomization in sim-to-real transfer [13], HAT for CIM faces an additional difficulty: device-to-device (D2D) mismatch is spatially structured and fixed for each hardware instance, unlike independent thermal noise. Standard HAT therefore tends to optimize against one sampled mismatch map. As we show below, this choice can make the model fit a particular hardware instance rather than the deployment distribution, leading to catastrophic accuracy collapse when transferring to fresh hardware.

System-level CIM simulators established effective workflows for inorganic memories. DNN +NeuroSim connected device models to peripheral-circuit costs and network accuracy [2]. MemTorch embedded memristive non-idealities into PyTorch for end-to-end evaluation [14], AIHWKIT exposed analog HAT and inference simulation in a practical software stack [15], and CrossSim provides a GPU-accelerated crossbar-accuracy workflow with parasitic resistance, ADC, drift, and lookup-table-based training models [16]. These frameworks remain anchored mainly to inorganic resistive memories, and they do not directly capture the coupled photoresponse, retention, and write nonlinearity that define organic optoelectronic inference.

At the same time, the broader organic-device literature provides enough empirical structure to motivate a deployment-facing simulator. Recent array demonstrations report energy-efficient organic device arrays, lithographically patterned near-infrared OPECT arrays, and large-scale optoelectronic synapse arrays for visual neural networks [8;9;17;18]. Related studies also point to active-matrix addressing, spatial optical response control, crosstalk, and thermal stability as practical system constraints rather than purely materials-level observations [19–23]. Taken together, this literature provides plausible ranges for conductance windows, state counts, retention behavior, and array-level artifacts in an organic-CIM workflow.

The mapping problem at the network level is likewise constrained by prior CIM practice. Dense linear operators are natural crossbar targets, whereas depthwise and highly irregular operators often suffer from poor utilization [24–26]. Recent heterogeneous CIM accelerators for Vision Transformers confirm this pattern: linear projections are mapped to analog arrays, while attention scores, softmax normalization, and dynamic interactions typically remain digital because of their precision requirements and irregular access patterns [27–29]. Low-bit ViT studies further show that attention-adjacent computations remain especially sensitive below 6 bits [30–32]. These observations motivate a hybrid analog–digital deployment model rather than an all-analog transformer implementation.

Here we present a behavioral simulation framework that bridges this materials-to-system gap. The framework incorporates organic-device-specific non-idealities—photoresponse, retention, and write nonlinearity—through a replaceable device-profile interface and applies a common mixed-signal deployment model to ResNet-18, ConvNeXt-Tiny, and Tiny-ViT-5M. Three constraints dominate in this setting. First, ADC resolution rather than nominal conductance quantization becomes the primary accuracy bottleneck: reducing ADC resolution below 6 bits causes abrupt collapse. Second, standard HAT overfits a fixed D2D realization and fails under fresh-instance transfer; Ensemble HAT, which resamples D2D masks at each training epoch, raises fresh-instance accuracy to $86.37 \pm 1.54\%$. Third, severe nonlinear write ($NL = 2.0$) remains difficult to recover under the present gradient-scaling approximation, limiting accuracy to $27.72 \pm 0.82\%$.

The framework is intentionally behavioral rather than circuit-accurate: array-level parasitics (IR drop, sneak paths), thermal effects, and pulse-faithful programming dynamics are not modeled explicitly and are revisited in Section 3. Its purpose is to identify which device characteristics most constrain edge-vision accuracy before full array hardware becomes available.

Table 1: Digital FP32 baselines for the evaluated backbones and datasets. Tiny-ViT CIFAR-10 reports the locked three-seed mean; the asterisk marks a single-run estimate for ConvNeXt–Flowers-102.

Dataset	ResNet-18	ConvNeXt-Tiny	Tiny-ViT-5M
CIFAR-10	95.46%	90.74%	98.06%
CIFAR-100	78.64%	64.12%	86.94%

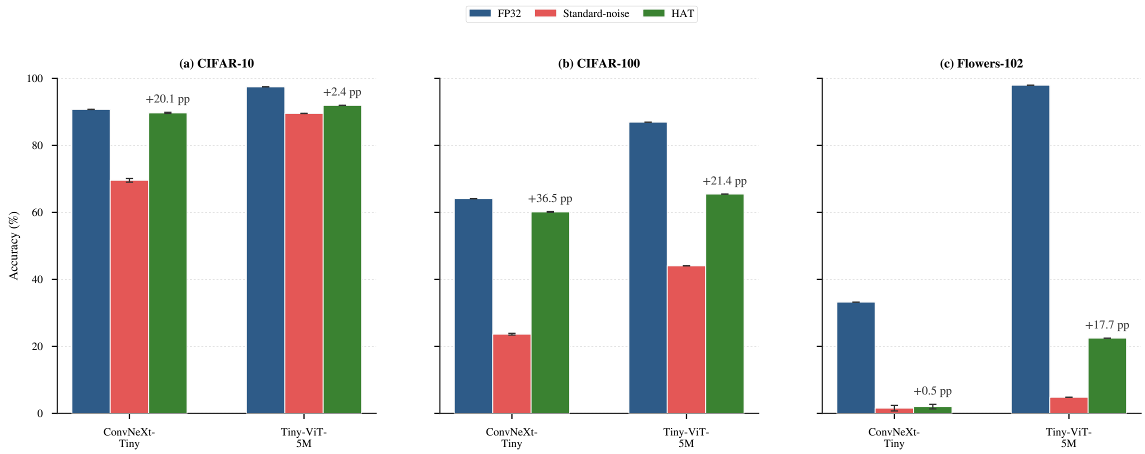

Figure 1: Cross-dataset accuracy under canonical deployment (4-bit quantization, 5% C2C, 10% D2D variability). Error bars denote ±1 standard deviation where Monte Carlo statistics are available; bars without visible error bars indicate deterministic baselines or currently available point estimates.

2 Results

2.1 Baseline Digital Performance

We establish digital FP32 baselines (Table 1). Tiny-ViT-5M reaches 98.06% on CIFAR-10 in the locked three-seed baseline, while ConvNeXt-Tiny trains from scratch to 90.74%. The gap widens on CIFAR-100 and Flowers-102, highlighting complementary testbeds: ConvNeXt probes analog adaptation from random initialization, whereas Tiny-ViT probes foundation-model deployment.

2.2 Quantization and Noise Resilience

Quantization alone introduces only a modest penalty in 4-bit hybrid models. The zero-noise hybrid control (V2) retains 97.39%, suggesting that 4-bit mapping by itself is not the dominant source of error in the present setup. Under the canonical uniform-noise profile, digital scale recovery suppresses much of the effective analog perturbation below the 4-bit least significant bit, so many noisy conductance realizations are mapped back into the same recovered weight bin after calibration. This behavior disappears once the noise law becomes state-dependent (§2.7).

Deployment fragility increases with task complexity (Figs. 1–2). On CIFAR-100, hardware-aware training improves Tiny-ViT from 44.06% to 65.48% and ConvNeXt from 23.86% to 60.54%. Results on Flowers-102 (Supplementary Table S6) show limited recovery, consistent with the noise-sensitivity pattern at fine-grained tasks; however, the zero-noise hybrid control (V2) reaches 91.30% (±0.00% over 10 runs), indicating that the failure is driven by noise-data interaction rather than by hybrid deployment alone. Because the ConvNeXt Flowers-102 digital baseline is only a single-run boundary estimate, we use that dataset primarily as a low-data stress test rather than as a stable cross-architecture ranking benchmark.

An extended ADC sweep over five bit-widths (4, 6, 8, 10, 12 bits; 10 Monte Carlo runs each) supports the same 6-bit threshold on ConvNeXt-Tiny: accuracy jumps from 48.4% at 4-bit (±16.2%) to 88.6% at 6-bit (±0.3%), saturating at 89.7% by 12-bit (Supplementary Fig. S3).

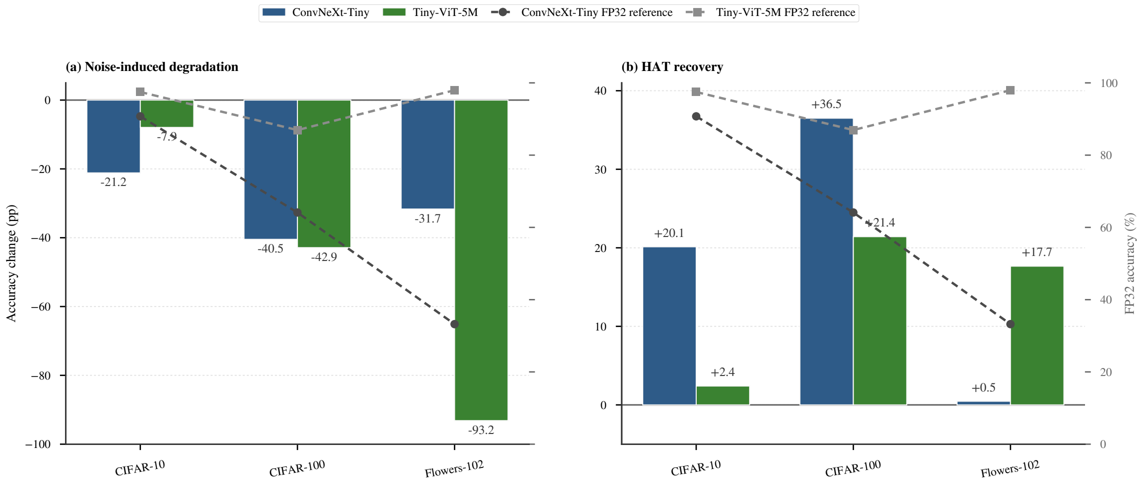

Figure 2: Accuracy degradation from FP32 to noisy deployment and HAT recovery. Tiny-ViT: 10-run MC; ConvNeXt: single-run estimates shown.

The iso-accuracy contour analysis (Section 2.6) further shows that this 6-bit cliff is largely independent of device variability, producing a consistent $\sim$7 pp transition across all tested σ_{D2D} levels. Readout precision emerges as a stronger constraint than static weight programming[30;31].

2.3 Retention and Temporal Drift

The corrected retention-aware follow-up model (V8) exhibits a plateau near **79% accuracy** through 10 000 s (Supplementary Fig. S6), consistent with the uniform decay model being adequate in the present regime.

2.4 Hardware-Instance Transferability

Experiments on fresh hardware instances (Supplementary Fig. S4) reveal an important limitation in transformer deployment. The standard fixed-mask HAT model (V4) collapses to chance level (10.00%) when evaluated on arrays with different fixed D2D realizations, indicating pronounced hardware-instance overfitting. This observation motivates the Ensemble HAT strategy discussed in Section 2.7.

2.5 Physical Front-end Compensation

Inverse-gamma compensation restores accuracy for high photoresponse non-linearity ($\gamma = 2.0$, +5.8 pp), but amplifies noise variance at high intensities—a design trade-off (Supplementary Figs. S7–S8).

2.6 Iso-Accuracy Operating Envelope

To characterize the joint tolerance of the Ensemble HAT model to device variability and readout precision, we perform a 63-point sweep over σ_{D2D} and ADC resolution. The grid spans $\sigma_{\mathrm{D2D}} \in \{1, 3, 5, 8, 10, 15, 20\}$% and ADC resolution $\in \{2, 3, 4, 5, 6, 7, 8, 10, 12\}$ bits. Each point is evaluated with 10 Monte Carlo runs at $\sigma_{\mathrm{C2C}} = 5\%$ and $NL = 1.0$. Figure 3 presents the resulting iso-accuracy contour map.

Three regimes emerge. Below 5-bit ADC, accuracy collapses to near-chance regardless of D2D magnitude. The transition from 5-bit to 6-bit yields a consistent $\sim$7 pp jump across all D2D levels, identifying 6-bit readout as the minimum viable precision. Above 6-bit, accuracy saturates and the remaining degradation is dominated by D2D variability: at $\sigma_{\mathrm{D2D}} \leq 10\%$ the model maintains >84%, while $\sigma_{\mathrm{D2D}} = 20\%$ reduces it to 71–77%.

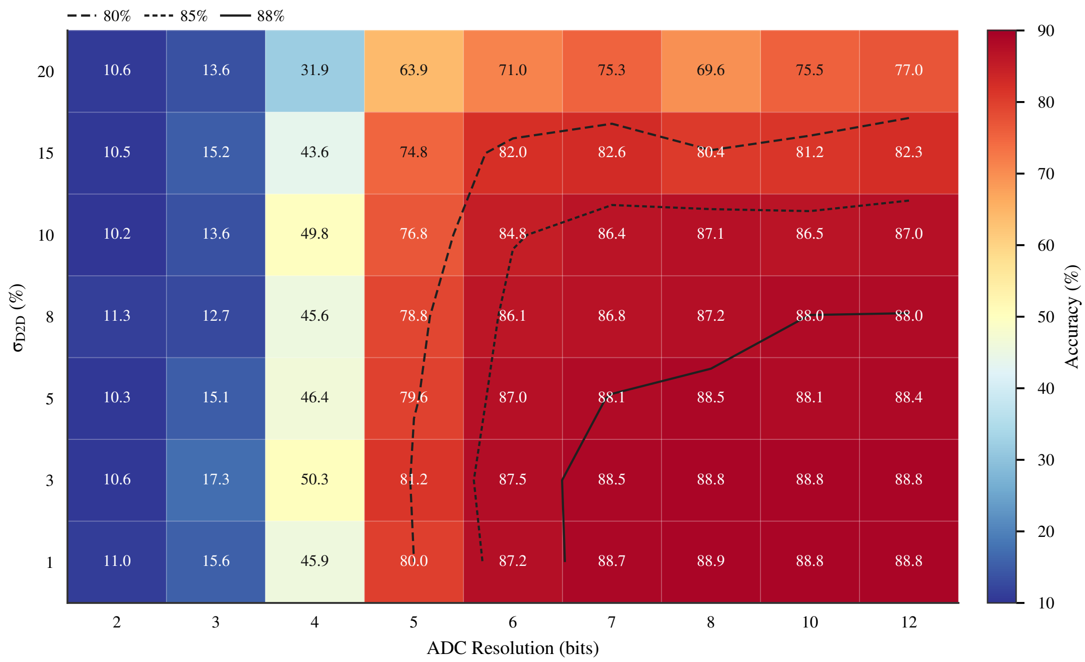

Figure 3: Iso-accuracy contour map for the Ensemble HAT Tiny-ViT model under joint σ_{D2D} and ADC precision sweep ($\sigma_{\mathrm{C2C}} = 5\%$, $NL = 1.0$, 10 MC runs per point). The 6-bit ADC cliff and the D2D-dominated degradation in the operational regime are visible.

A first-order Sobol sensitivity analysis supports this two-phase structure. Over the full parameter space the ADC index $S_{\mathrm{ADC}} = 0.98$, reflecting the sub-6-bit collapse. Within the operational region ($\geq$6-bit, $\sigma_{\mathrm{D2D}} \leq 15\%$), the D2D index $S_{\mathrm{D2D}} = 0.92$, indicating that device variability becomes the binding constraint once minimum readout precision is satisfied. The interaction term is negligible in both regimes (<4%), indicating that ADC and D2D impairments decouple to first order.

2.7 Non-Linear Writing and Hardware-Aware Training

Stress testing HAT recovery under richer device laws reveals a strongly regime-dependent outcome. Under matched proportional-noise HAT, Tiny-ViT recovers to 97.37 $\pm$ 0.05%, whereas severe write non-linearity ($NL = 2.0$) limits recovery to 27.72 $\pm$ 0.82% under the present gradient-scaling approximation. For ConvNeXt-Tiny, proportional-noise HAT reaches 91.98% in the best individual run, but the corresponding three-seed reproducibility evaluation yields **84.75 $\pm$ 0.72%**. The gap between the best run and the cross-seed mean indicates notable seed sensitivity, underscoring the importance of reporting reproducible analog HAT results rather than isolated favorable runs. Resampling D2D masks during training alleviates instance overfitting, preserving 86.37 $\pm$ 1.54% across 10 fresh arrays. The accompanying frequency ablation (Supplementary Fig. S4) indicates that per-epoch resampling performs best in the present setup: epoch-level resampling yields 88.41%, compared with 87.76% (every 20 epochs), 87.31% (every 5), 87.18% (fixed at initialization), and 86.16% (per-batch). The single-epoch cadence balances sufficient exposure to diverse mismatch maps against training stability, while leaving overall training time effectively unchanged in our logged runs. Figure 4 illustrates the distinction: standard HAT trains against a single fixed D2D realization, whereas Ensemble HAT resamples D2D masks each epoch to target the mismatch structure encountered across distinct hardware instances.

The accuracy results are summarized in Table 2. Tiny-ViT demonstrates superior baseline accuracy and significant recovery under HAT, especially on complex tasks, though it remains sensitive to nonlinear write dynamics.

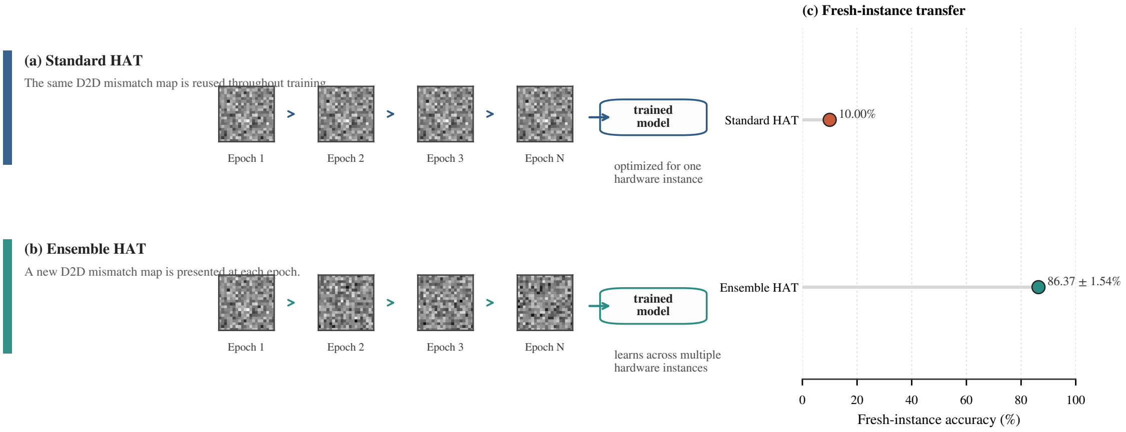

Figure 4: Ensemble HAT concept. Standard HAT reuses one fixed D2D mismatch map throughout training, whereas Ensemble HAT resamples the mismatch map each epoch. The fresh-instance transfer panel highlights the resulting generalization gap between the two training protocols.

Table 2: Summary of classification accuracy (%) across architectures and physical regimes. Cross-dataset deployment rows report locked best-checkpoint values; entries with ± denote stochastic physical-extension evaluations. The asterisk marks the ConvNeXt–Flowers-102 single-run baseline.

Architecture	Regime	CIFAR-10	CIFAR-100
ResNet-18	FP32 Digital	95.46	78.64
ResNet-18	Standard noisy deployment (R3)	16.48	1.00
ResNet-18	Uniform-noise HAT (R4)	90.37	1.00
Tiny-ViT-5M	FP32 Digital	98.06	86.94
ConvNeXt-Tiny	FP32 Digital	90.74	64.12
ConvNeXt-Tiny	Standard noisy deployment (C3)	70.48	23.86
ConvNeXt-Tiny	Uniform-noise HAT (C4)	89.91	60.54
ConvNeXt-Tiny	Proportional-noise HAT (3-seed extension)	-	84.75 ± 0.72
Tiny-ViT-5M	Standard noisy deployment (V3)	89.54	44.06
Tiny-ViT-5M	Uniform-noise HAT (V4)	91.94	65.48
Tiny-ViT-5M	Front-end compensation (V6)	95.82 ± 0.12	-

2.8 Case Study: Zero-Shot Transfer to a Literature-Calibrated Device

To provide a literature-anchored reference point against reported device data, we conducted a case study simulating zero-shot transfer to a recent 2025 OPECT array calibrated from literature[9] (conductance range 47.3, 34 states). Using noise parameters derived from the reported characteristics ($\sigma_{\mathrm{C2C}} = 2\%, \sigma_{\mathrm{D2D}} = 3\%$), the standard HAT model collapses to 10.00% on this unseen hardware profile. In contrast, the Ensemble HAT model successfully maintains **88.53%** accuracy (Figure 5).

This literature-anchored case study illustrates how the framework links reported physical metrics to algorithm deployment risk. The stark contrast between standard and Ensemble HAT on literature-calibrated benchmarks suggests that addressing spatial mismatch map overfitting is important for realizing the potential of emerging organic optoelectronic arrays.

3 Discussion

3.1 Principal Accuracy Bottlenecks

The combined ResNet, ConvNeXt, and Tiny-ViT analyses indicate that the dominant limitations under the canonical deployment regime (standard 4-bit quantization with 5% C2C

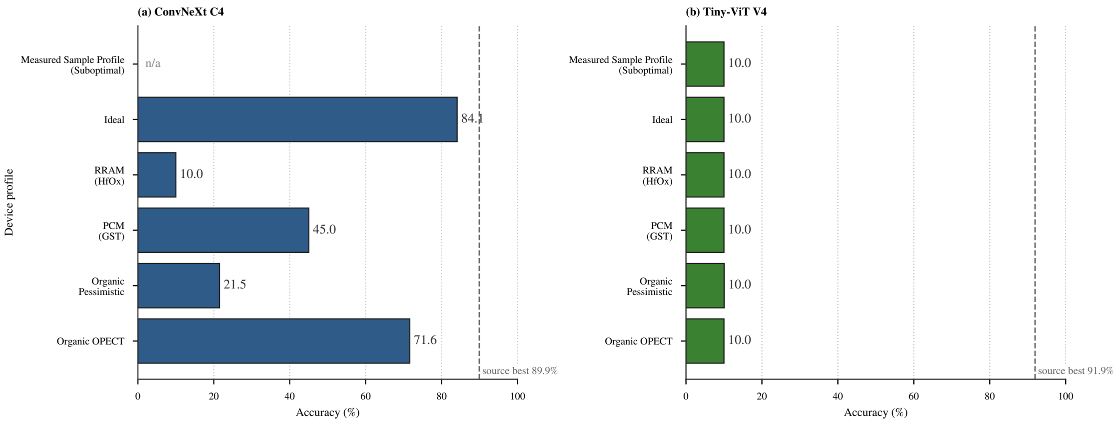

Figure 5: Zero-shot transfer across alternative device profiles. Tiny-ViT trained with standard HAT collapses to chance level (10.00%) when evaluated on literature-calibrated or measured profiles (OPECT 2025[9], PCM[11], RRAM[2], and a measured suboptimal organic sample profile), whereas Ensemble HAT recovers $> 80\%$ accuracy across the profiled regimes.

and 10% D2D variability) are not the ones most often emphasized at the device level. Once hardware-aware training (HAT) and scale recovery are available, standard cycle-to-cycle (C2C) and device-to-device (D2D) variability do not dominate the error budget. Three constraints dominate instead.

A quantitative decomposition supports this ordering. First-order Sobol indices computed from a 7×9 D2D–ADC grid show that, over the full parameter space, ADC precision accounts for 97.6% of accuracy variance ($S_{\mathrm{ADC}} = 0.98$), while D2D contributes only 1.8%. However, conditioning on the operational envelope ($\geq$6-bit ADC, $\sigma_{\mathrm{D2D}} \leq 15\%$), the hierarchy inverts: D2D variability accounts for 92.2% of residual variance ($S_{\mathrm{D2D}} = 0.92$). This two-phase structure has a direct hardware implication: designers should first secure 6-bit readout, then invest the remaining error budget in reducing device-to-device mismatch.

The first is ADC resolution. The transition around 6-bit ADC resolution marks a clear change in model behavior: below this point transformer inference degrades substantially, indicating that improvements in conductance control remain gated by readout precision in the present mixed-signal stack.

The second is hardware-instance overfitting. The collapse of V4 under fresh-instance transfer (10.00% accuracy) shows that standard HAT readily fits a single fixed D2D realization. Resampling D2D masks during training raises fresh-instance accuracy to $86.37 \pm 1.54\%$, indicating that multi-instance-aware training can mitigate this transfer failure mode. Mechanistically, Ensemble HAT changes the fixed spatial mismatch map itself across epochs rather than merely perturbing one nominal instance.

A residual gap to the 98.06% digital baseline persists under matched deployment conditions. Part of this difference may reflect regularization from per-epoch D2D resampling rather than analog noise alone. Capacity-aware Ensemble HAT schedules remain an open optimization problem.

The third is the scale-masking regime observed in V2. The high accuracy of V2 under standard uniform noise is a conditional property of the canonical simulator configuration rather than a universal form of robustness. When digital scale recovery maps analog perturbations below the 4-bit quantization step, many realizations remain in the same recovered weight bin after calibration. This protection depends on accurate post-array gain calibration and disappears once the noise law becomes state-dependent, as shown by the proportional-noise experiments.

3.2 Transformer Sensitivity to Non-Idealities

Comparing the dual testbeds—ConvNeXt (CNN) and Tiny-ViT (Transformer)—reveals that the Transformer architecture is more fragile under the present physical stress tests. Because the two models serve complementary roles (ConvNeXt from scratch versus Tiny-ViT from pre-trained fine-tuning), this result should be read as evidence of a strong architecture-and-training interaction rather than as a universal ranking of all Transformers against all CNNs.

The proportional-noise experiments illustrate this most clearly. The collapse of V4 under proportional noise (10.00% accuracy), contrasted with the relative stability of ConvNeXt under moderate non-linearities, indicates that Transformer inference depends on the precision of token projections. Because attention maps arise from global interactions, perturbations at a small number of high-conductance devices can disrupt the spatial weighting of the entire block. The optical front-end experiments point in the same direction: the larger accuracy drop in V6 than in ResNet R4 implies that input distortions are propagated more readily through sequence-wide attention than through a purely convolutional hierarchy.

Robustness is highly regime-matched. Tiny-ViT can recover strongly when trained and evaluated under the same proportional-noise law, yet the resulting checkpoint does not transfer back to the canonical uniform-noise regime. In this framework, robustness is distribution-specific rather than noise-invariant.

The framework remains reasonably resilient overall. A compound stress test combining all modeled non-idealities (2% C2C, 3% D2D, 6-bit ADC, 1% IR drop, 1% sneak path, and 1,000 s retention) maintained an 89.61% accuracy for Tiny-ViT, suggesting that the mixed-precision architecture with Ensemble HAT can tolerate moderate cumulative deployment stress within the present model.

3.3 Task Complexity and Data Starvation

The multi-dataset results show a clear interaction between task difficulty and analog robustness. HAT recovery is most valuable on CIFAR-100, where the decision boundaries are comparatively fine, whereas CIFAR-10 is sufficiently simple that even noisy standard models remain functional. However, ResNet-18 exhibits a catastrophic failure mode on CIFAR-100 under the canonical noisy regime, collapsing to chance-level accuracy (1.00%) regardless of HAT training. A post hoc audit showed that the earlier CIFAR-10 inconsistency arose from a legacy model-loading issue rather than from the analog inference path itself. Replay with the compatibility loader recovered the two affected CIFAR-10 checkpoints to 94.12% and 89.60%. Table 2 retains the original checkpoint-best convention for the canonical R4 row (90.37%), whereas the discussion quotes the post-fix replay value because that audit established that the saved weights were valid and that the earlier collapse was a load-time artifact. The CIFAR-100 result, by contrast, remained at 1.00% after this correction and was therefore confirmed as a genuine optimization failure rather than a load-time artifact. Because this failure reflects a limitation of the present ResNet conversion and training recipe rather than a robust cross-architecture physical trend, we do not draw architectural conclusions from the ResNet-18 CIFAR-100 result and focus the cross-architecture comparison on Tiny-ViT and ConvNeXt, which both pass baseline sanity checks under the same workflow. Flowers-102 marks the opposite extreme for the current recipe: Tiny-ViT recovers only partially, and the ConvNeXt control remains unstable in this low-data regime. Because the ConvNeXt Flowers-102 baseline is only a single-run boundary estimate, we treat that dataset as a qualitative low-data stress test rather than as a stable cross-architecture ranking benchmark.

3.4 The "Analog Ceiling" and Energy Efficiency

A first-order analytical energy model projects a potential 11.45× reduction in dense-projection energy relative to FP32 digital inference, though this illustrative upper bound rests on unvalidated edge-node placeholders. Moderate routing overhead (10–50%) reduces this gain to 9.90–11.10×, and digital execution of dynamic attention still dominates the overall footprint (57.9% of total energy). Even so, the estimated 273.94 μJ footprint of the hybrid stack compares favorably with reported INT8 ViT digital accelerators[33], suggesting that hybrid deployment may offer an efficiency advantage in favorable regimes. These numbers should be interpreted as first-order system-level upper bounds under the current placeholder constants rather than as chip-predictive routed-silicon estimates.

3.5 Limitations

Several physical idealizations of the behavioral model should be noted. The energy model parameters (E_{cell}, E_{ADC}, E_{DAC}) are analytical placeholders rather than chip-predictive measured values. Adding extra routing overhead equal to 10%, 30%, or 50% of the analog-MAC budget would reduce the referenced gain, indicating that the qualitative advantage is robust but sensitive to interconnect assumptions.

Regarding array non-idealities, our sensitivity sweep for position-dependent IR drop (up to 3%) and sneak-path currents (up to 2%) suggests robustness to moderate parasitic effects. However, these baseline ranges are derived from metallic-interconnect ReRAM literature[34;35]. Given that organic semiconductor arrays often exhibit significantly higher sheet resistances, this range should be interpreted as a lower-bound sensitivity probe rather than a prediction of full-scale array behavior. Future integration of measured organic-array sheet-resistance data will be necessary to refine these bounds. At the same time, the accompanying supplementary proxy-sensitivity sweeps and parameter-risk tables indicate that, within the tested uncertainty ranges, proxy-value perturbations do not reverse the main ranking of ADC-limited versus D2D-limited regimes. Furthermore, the observed "scale-masking" effect, where small C2C noise is absorbed within quantization bins, relies heavily on the assumption of ideal calibrated digital rescaling. A full-test hook-based check reported in the Supplementary Information shows that moderate ADC calibration errors (±0.5 LSB offset, ±5% gain error, 0.5 LSB INL) change the canonical Tiny-ViT V4 accuracy by only 0.18 percentage points, whereas a deliberately pessimistic combination causes a 5.14 percentage-point drop. In practical deployment, readout-chain non-idealities therefore remain a gate-keeping condition, but the protected regime is not immediately destroyed by moderate calibration error.

Finally, the observed degradation at $NL = 2.0$ ($27.72 \pm 0.82\%$) should be interpreted as a recovery limit for the present gradient-scaling surrogate and training recipe, rather than as an immutable physical constraint of organic optoelectronic materials themselves. A supplementary group-wise gradient diagnostic on the frozen Tiny-ViT V4 checkpoint further localizes this surrogate failure: under matched forward conditions, enabling $NL = 2.0$ only in the MLP analog blocks lowers the affected-parameter gradient cosine to 0.815 and the norm ratio to 0.671, whereas patch embedding and attention projections remain essentially unchanged. This indicates that the present nonlinear-write bottleneck is concentrated in the MLP path rather than being uniformly distributed across the transformer. Alternative straight-through formulations or specialized non-linear-aware optimizers may shift this boundary and remain an open research direction. Taken together, these simplifications mean that the framework should be read as a materials-to-system decision aid for ranking deployment risks under plausible parameter ranges, not as a chip-predictive emulator for a specific fabricated array.

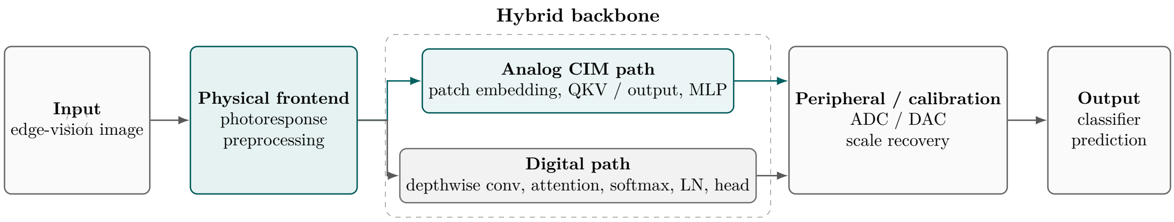

Figure 6: Hybrid organic-CIM inference stack. Dense weight-stationary operators are executed on analog crossbars, whereas dynamic attention and control-heavy operators remain in the digital path.

3.6 Outlook

An immediate next step is to close the loop between laboratory characterization and simulation through automated profile fitting. Multi-instance HAT schedules should also be explored further, particularly where transferability and capacity appear to compete. A preliminary cross-framework comparison against CrossSim shows reasonably consistent baseline inference under clean conditions (86.2% ours vs. 83.7% CrossSim on a 1 000-sample subset at 8-bit ADC), but the accuracy gap widens substantially under noise injection (81.6% vs. 67.2% at $\sigma = 5\%$), highlighting the sensitivity of accuracy predictions to the specific noise-to-conductance mapping. Because the two toolchains use different internal approximations for mapping device assumptions to effective conductances, this discrepancy should be read as a joint-calibration problem rather than as evidence that one simulator is categorically correct and the other is not. These shared-regime sanity checks can be extended into broader numerical-equivalence studies that also include retention, photoresponse, and richer write dynamics.

4 Methodology

4.1 Hybrid Analog/Digital Deployment

We adopt a hybrid analog/digital inference stack: dense linear operators execute on differential-pair crossbars; control-heavy operators remain digital. Figure 6 summarizes this division at the system level, and Table 3 lists the operator-level rationale following established practice in heterogeneous CIM accelerators[24–28].

Under this mapping, 87.7% of parameters are assigned to analog execution, whereas 57.9% of the estimated energy remains in the digital domain, largely because of attention operations. This split reflects a basic architectural constraint: analog crossbars are well suited to weight-stationary matrix multiplication, but much less suited to dynamic, input-dependent computations[27]. Recent heterogeneous CIM accelerators for Vision Transformers follow the same logic, keeping attention digital even as linear projections move to analog arrays[25;26;28]. The present organic device literature likewise demonstrates regular static array programming more naturally than fully dynamic interaction engines[8;9].

4.2 Modeling Physical Non-Idealities

The framework injects quantization, cycle-to-cycle (C2C) and device-to-device (D2D) variability, ADC distortion, retention drift, and nonlinear-write surrogates directly into the forward path. Each analog-mapped weight W_{ij} is converted into a differential conductance representation quantized to n_{states} levels. We first decompose the weight into positive and negative branches:

$$W_{ij}^{+} = \max(0, W_{ij}), \quad W_{ij}^{-} = \max(0, -W_{ij}) \tag{1}$$

These are then mapped to the physical conductance window $[G_{\min}, G_{\max}]$:

$$G_{pos,ij} = G_{\min} + \frac{W_{ij}^{+}}{\max|W|}(G_{\max} - G_{\min}) \tag{2}$$

Table 3: Analog/digital operator mapping rationale. The split maximizes array utilization for weight-stationary operations while preserving precision for dynamic, irregular computations.

Operator	Domain	Rationale
Patch embedding (conv)	Analog	High weight reuse, regular access pattern; natural fit for crossbar MVM
QKV projections	Analog	Dense matrix-vector multiplications; weights stationary during inference
Attention output proj.	Analog	Post-attention linear projection; same characteristics as QKV
Feed-forward networks	Analog	MLP layers dominate parameter count; efficient crossbar utilization
Depthwise convolutions	Digital	Low weight reuse, irregular spatial access; poor crossbar utilization [24]
Layer normalization	Digital	Requires mean/variance computation across features; hard to parallelize on crossbar
$\mathbf{QK}^\top$ (attention scores)	Digital	Dynamic, input-dependent computation; irregular memory access [28]
Softmax	Digital	Non-linear transcendental function; precision-critical; no efficient analog implementation [29]

with $G_{neg,ij}$ calculated symmetrically. The differential-pair scheme enables signed-weight representation while partially suppressing common-mode noise [36;37]. Hardware-aware training (HAT) is implemented by keeping these forward perturbations active during optimization while propagating gradients through a straight-through estimator. For transformer deployments, the recovered post-array weight scale is modeled as an ideal calibrated digital rescaling step with fixed layer-wise factors, and its control overhead is therefore not itemized separately in the energy model.

First-order modeling of nonlinear write (`NL_LTP/NL_LTD`) and double-exponential retention is supported, with parameters anchored to measured DNTT transients [38]. For optoelectronic sensing, we incorporate a physical frontend model (V6) characterized by the photoresponse exponent γ_{phys}, representing the power-law relation between light intensity and conductance, and the dark current I_{dark}, which sets the dynamic range floor. The released profiler uses representative edge-node energy and latency parameters.

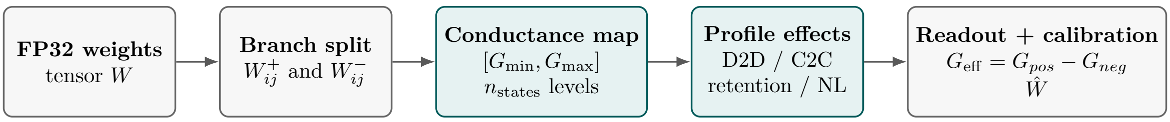

Figure 7: Behavioral weight-to-conductance mapping. Floating-point weights are split into differential branches, mapped to a bounded conductance window, quantized, and perturbed by the selected device profile before differential readout.

4.3 Profile-Driven Substitution Interface

The profile interface allows technology-specific parameters ($\sigma_{\text{D2D}}, \sigma_{\text{C2C}}, n_{\text{states}}, G_{\max}/G_{\min}$) to be substituted directly into the simulation workflow. In practice, this means that the same framework can absorb literature priors at present and measured-device characterizations later without changing the underlying evaluation pipeline. We therefore archive the optimization settings and random seeds needed for independent reruns. The resulting methodology is intended as a first-order behavioral framework for deployment-facing risk assessment, rather than as a pulse-accurate emulator.

Table 4: Notation used for the Tiny-ViT experiment family in the main text.

ID	Meaning
V1	FP32 digital baseline
V2	Quantized hybrid control without analog noise
V3	Standard training with a fixed D2D realization; noisy deployment baseline
V4	Uniform-noise hardware-aware training result used as the canonical deployment model
V5	Pessimistic robustness stress test with larger uniform noise
V6	V4 plus physical front-end compensation
V7	Legacy retention-aware model, excluded from the canonical claims
V8	Corrected retention-aware follow-up used only for retention checks

5 Experimental Setup

We evaluate ResNet-18 on CIFAR-10 as an entry validation platform, ConvNeXt-Tiny trained from scratch as the higher-capacity CNN testbed, and Tiny-ViT-5M fine-tuned from ImageNet as the deployment-oriented transformer setting. CIFAR-10, CIFAR-100, and Flowers-102 provide a progression from easier to harder classification settings and from relatively data-rich to data-limited regimes.

The experiments fall into three groups: ResNet baseline quantization and front-end trends, ConvNeXt retention and proportional-noise studies, and Tiny-ViT hardware-aware training together with physical-extension tests such as nonlinear write and fresh-instance transfer. Detailed optimization settings and the Monte Carlo evaluation protocol are summarized in the Supplementary Information.

For ConvNeXt, C1–C4 denote the analogous FP32, quantized/no-noise, noisy-baseline, and hardware-aware training runs; C4 also anchors the proportional-noise and retention follow-up studies described in the Supplementary Information.

Parameter provenance is summarized in the Supplementary Information, together with the optimization settings and evaluation conditions used for the reported runs. The present study therefore targets reproducible reruns through archived configurations, logs, and model lineage. We also emphasize that the framework operates at the simulation level: all device parameters are literature-derived or proxy-calibrated rather than extracted from fabricated devices under direct measurement. Validation against physical hardware remains for future work.

6 Conclusion

We presented a first-order behavioral simulation framework for organic optoelectronic compute-in-memory inference that links materials-level characterization to task-level vision performance. By combining differential mapping, variability, retention, and peripheral constraints within a profile-driven workflow, the framework serves as a practical materials-to-system decision aid for evaluating deployment risk before full array hardware becomes available.

Several observations stand out. A 63-point iso-accuracy sweep reveals a two-phase constraint hierarchy: ADC resolution below 6 bits causes abrupt collapse ($S_{\mathrm{ADC}} = 0.98$ over the full grid), while within the operational envelope device-to-device variability becomes the binding factor ($S_{\mathrm{D2D}} = 0.92$). Fixed hardware-instance transfer, rather than nominal quantization, is the most prominent deployment risk in the present regime: standard HAT collapses on fresh arrays, whereas Ensemble HAT raises fresh-instance accuracy from 10.00 % to $86.37 \pm 1.54\%$. In the canonical uniform-noise setting, quantization and nominal stochasticity are less limiting once scale recovery is active. By contrast, severe write non-linearity remains difficult to recover under the present gradient-scaling approximation ($27.72 \pm 0.82\%$), and determining whether physical organic devices enter this regime will require direct nonlinear-write measurements.

The framework translates reported device behavior into deployment-facing system risk through an explicit profile interface. Under a literature-anchored OPECT case study, zero-shot deployment reaches 88.53% without any change to the evaluation workflow. The same structure should simplify future measured-device calibration while preserving model-level comparability within a common workflow.

Data Availability

All datasets used in this study (CIFAR-10, CIFAR-100, and Flowers-102) are publicly available through standard PyTorch torchvision and HuggingFace datasets libraries. Source-data tables for the plotted figures, together with the submission-matched logs and inference traces, will be deposited in a reviewer-accessible repository at submission and released publicly at acceptance.

Code Availability

The simulation framework, training and evaluation scripts, device-profile utilities, and plotting code will be provided to editors and reviewers through a reviewer-accessible archive at submission. A curated public release will be made available at https://github.com/OrgOptEdge/compute_vit upon acceptance.

Competing Interests

The authors declare no competing interests.

Author Contributions

S.L. conceived the framework, implemented the simulator, conducted all experiments, and wrote the manuscript.

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